Tubular Daylight Collector Dome - Expanded Prototype Guide

Detailed, easier-to-read companion to the 14-inch / 356 mm collector dome engineering package.

1	2	3	4	5
UNDERSTAND	PLAN	FABRICATE	INSPECT	VALIDATE

DOCUMENT STATUS	Preliminary prototype design only. Wind, snow, hail, impact, fire, flashing, leakage, structural attachment, and local code compliance require qualified project-specific review before outdoor installation.

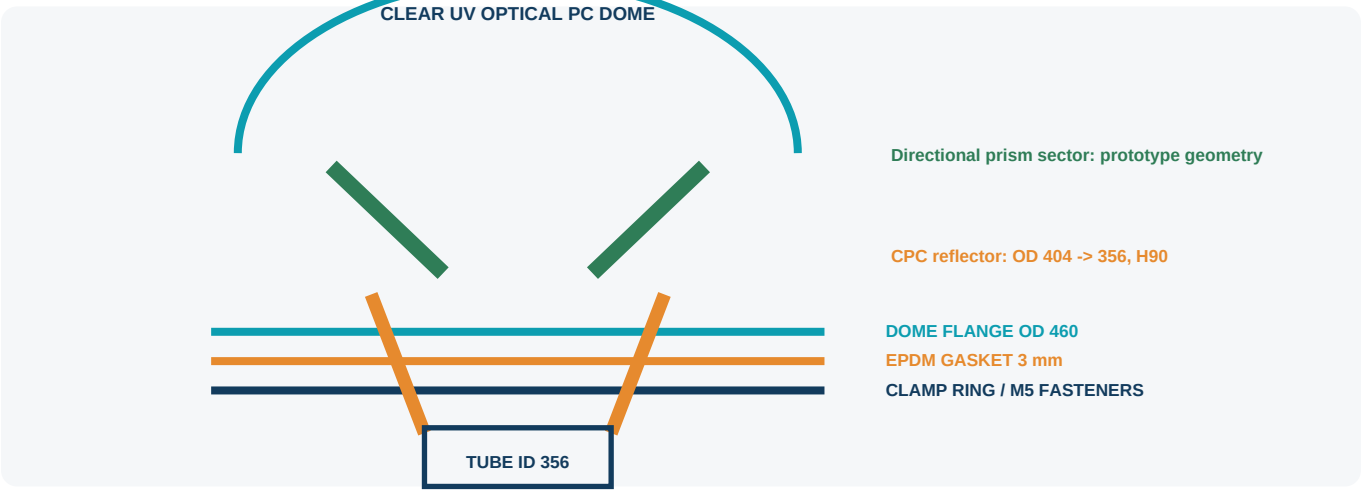
What this expanded edition adds

- Plain-language explanation of every major interface and why each controlling dimension matters.
- Simplified diagrams for the optical path, joints, supports, seals, routing, and service interfaces.
- Step-by-step fabrication and installation travelers with CHECK and HOLD gates.
- Inspection logic, common failure clues, commissioning records, and prototype release sign-off.
- The original engineering package is appended unchanged and remains the controlling dimensional source.

HOW TO USE THIS DOCUMENT	Read the explanatory supplement first. Use the original sheets for fabrication geometry and exact source values. Do not treat a plain-language explanation or example calculation as a new released design value.

1. Collector assembly - what each layer does

The collector combines weather enclosure, broad-angle daylight admission, directional low-sun redirection, reflective coupling, and a sealed mechanical interface to the 356 mm tube.



ELEMENT	SOURCE DESIGN	FUNCTION
Clear dome	Outer base Dia. 420; rise 230; flange Dia. 460	Weather enclosure + broad-angle optical entrance
Directional prism sector	120 deg azimuth; meridian 5-55 deg; prototype 35/55 facets	Redirect selected low-altitude rays downward
CPC transition	Dia. 404 -> 356 x H90; protected silver target >=98%	Guide admitted edge rays into tube
Gasket / clamp	3 mm EPDM; 16 M5 positions on PCD 430	Seal while allowing polymer movement
Tube collar	ID 356 x 100 engagement	Concentric transition to tube system

PROTOTYPE OPTICS	The prism dimensions are not a released final optical prescription. The package requires ray tracing plus physical directional-light tests before geometry is frozen.

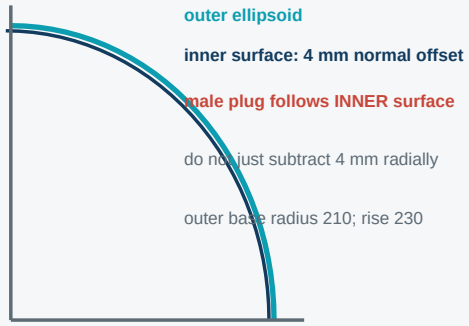
2. How to read the dome package

These terms appear repeatedly in the drawings, tooling notes, and validation plan.

TERM	PLAIN-LANGUAGE MEANING
Ellipsoid of revolution	A smooth 3D dome formed by rotating the 2D meridian equation around the vertical axis.
Meridian	A vertical slice through the dome center used to define the curved profile.
Normal offset	Wall thickness measured perpendicular to the curved surface. The inner surface is 4.0 mm inward along the normal, not simply 4 mm less radius.
Male plug	The forming tool that controls the inner surface of the thermoformed dome.
Prism sector	An asymmetric optical insert covering only a defined azimuth/meridian region to favor low-sun redirection.
CPC transition	Reflective non-imaging transition that guides edge rays from Dia. 404 to the Dia. 356 tube opening.
PCD	Pitch circle diameter for equally spaced fastener holes.
Stop sleeve	Spacer around a fastener that limits gasket/flange compression and reduces point loading.
Ray-trace starting point	Geometry is intentionally provisional until simulated and measured optical results correlate.
Release gate	A test/approval condition that must be satisfied before a design feature is frozen or installed permanently.

DRAWING STATUS	The drawings are fabrication-development documents. Printed dimensions and source tables control; sketches are explanatory and may be NTS.

3. Dome geometry and forming-tool definition



Outer surface equation

(r / 210)^2 + (z / 230)^2 = 1 for 0 <= z <= 230 mm. The outer base radius is 210 mm and the rise is 230 mm.

CRITICAL TOOLING RULE	The male plug follows the INNER surface, defined as a 4.0 mm normal offset from the outer optical surface. Do not create the tool by simply subtracting 4 mm from the radial coordinate.

MERIDIAN deg	OUTER r	OUTER z	INNER r	INNER z
0	210.00	0.00	206.00	0.00
10	206.81	39.94	202.86	39.30
20	197.34	78.66	193.54	77.40
30	181.87	115.00	178.33	113.13
40	160.87	147.84	157.69	145.41
50	134.99	176.19	132.28	173.25
60	105.00	199.19	102.86	195.81
70	71.82	216.13	70.34	212.41
80	36.47	226.51	35.71	222.58
90	0.00	230.00	0.00	226.00

4. Tool and finished-part tolerances - inspection meaning

CONTROL	SOURCE LIMIT	WHY / HOW TO CHECK
Optical profile	+/-0.50 mm over optical surface	Check against CMM/template/radial height points
Local waviness	<=0.25 mm over 50 mm chord	Prevents local optical distortion/scatter
Base cut plane	flat within 0.50 mm	Required for even gasket/clamp sealing
Flange OD	Dia. 460 +/-0.50	Matches clamp/mounting interface
Finished wall	target 4.0; minimum 3.0 at apex/transition	Reject thin regions; retain wall-thickness map
Optical finish	Polished tool surface; no machining marks	Tool defects print into optical surface
Shrink allowance	0.6% only as initial trial value	Measure actual resin/process and correct tool after prototypes

Recommended prototype measurement record

FIELD	ENTRY
Part ID	_____
Sheet lot / grade	_____
Starting thickness	_____
Min finished wall	_____
Max profile deviation	_____
Flange flatness	_____
Flange OD	_____
Forming method / cycle ref	_____

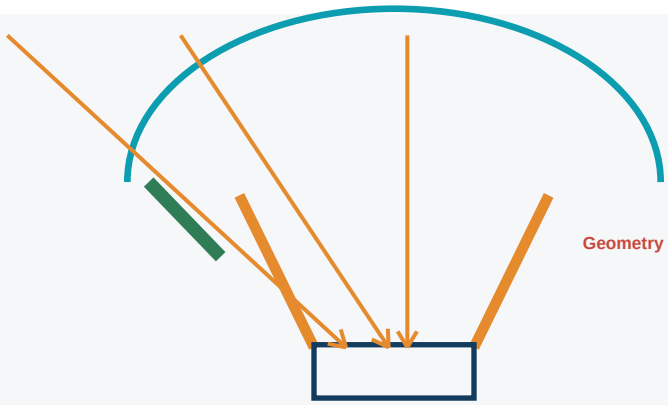
FREEZE TOOL ONLY AFTER REPEATABILITY	The validation plan says to freeze the dome tool only after at least three measured parts show repeatable profile and wall distribution.

5. Thermoforming process - step-by-step traveler

#	OPERATION	CONTROL / EVIDENCE
1	Build male plug	Use inner-profile coordinate table; sealed MDF may be used for trials, aluminum for repeat production.
2	Finish tool	Seal/polish optical surface; one Dia. 0.8 apex vent + twenty Dia. 0.8 base vents to vacuum.
3	Condition sheet	Use selected UV optical PC. Predry per supplier; 120 C for 2-3 h is only a source starting trial.
4	Thermoform	Pressure or vacuum form in supplier process window; heat uniformly; record temperatures/draw ratio.
5	Cool on tool	Hold until dimensionally stable; early release can cause springback, warpage, flange distortion.
6	Inspect before trim	Record profile and wall-thickness distribution; inspect optical defects.
7	Trim + drill	Trim flange Dia. 460; drill Dia. 7.0 polymer holes on PCD Dia. 430 with sharp tools/backing.
8	Install prism + CPC	Keep optical surfaces clean; avoid trapped dust, wrinkles, adhesive squeeze-out.
9	Seal + clamp	EPDM gasket, clamp ring, stop sleeves, bonded washers, M5 fasteners; gradual star tightening.
10	Validate	Leak, thermal, impact/hail, optical, directional, environmental tests before release.

SUPPLIER PROCESS OVERRIDES GENERIC START VALUES	The exact polycarbonate grade and forming equipment determine drying and forming windows. The source temperatures are development starting points, not universal settings.

6. Optical system - how the dome, prism, and CPC work together



Smooth dome: broad-angle transmission

Prism sector: selected low-angle redirection

CPC: couples edge rays toward tube

Geometry remains prototype until ray trace + test correlation

LIGHT CONDITION	DOMINANT ELEMENT	DESIGN INTENT
Diffuse sky	Smooth clear dome	Transmit broad angular range into collector/tube
High-angle direct sun	Primarily clear dome / central path	Admit useful light without unnecessary redirection
Selected low-angle sun	Directional prism sector	Refract selected rays downward rather than crossing/escaping
Edge rays	CPC reflective transition	Couple rays from Dia. 404 entrance toward Dia. 356 tube

PERFORMANCE METRIC	Evaluate useful light at the diffuser, not only peak noon illuminance. The source favors a wider, more uniform acceptance band over a narrow optical peak.

7. Prism sector geometry and orientation - prototype only

Starting prism geometry: 120-degree azimuth sector, meridian band 5 to 55 degrees, pitch 2.50 mm, depth 1.17 mm, 35/55-degree facets, and 1.0 mm base film. These values are controlled starting geometry for ray tracing and prototype trials, not final prescription.

CONDITION	ORIENTATION RULE
Northern Hemisphere	Face sector generally south / toward equator
Southern Hemisphere	Face sector generally north / toward equator
Final site azimuth	Choose using latitude, roof pitch, surrounding shading, and hours when useful daylight is most valuable
Permanent mark	Mark equator-facing direction on clamp ring so prototype orientation is repeatable

Variables that must enter the final optical optimization

- final resin refractive index / dispersion
- actual formed wall thickness
- site latitude and weather data
- roof orientation and pitch
- tube reflectance and run geometry
- diffuser distribution and glare target
- prism pitch, facet angles, sector width, and meridian coverage

HOLD - PRISM FREEZE	Freeze prism geometry only after ray tracing is correlated with physical directional-light tests.

8. CPC reflective transition - developed blank and build control

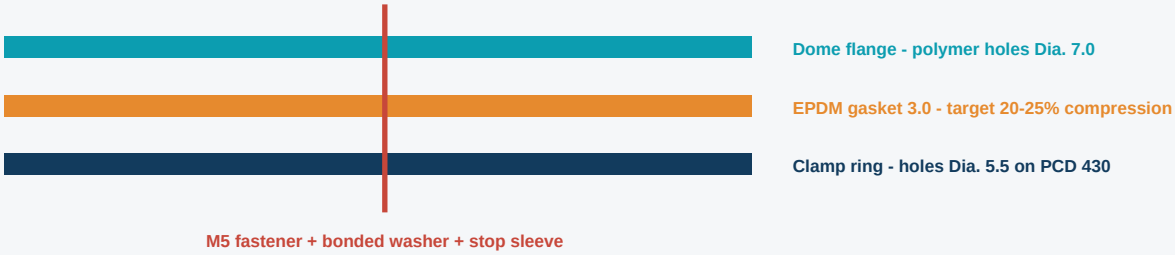
The CPC-like conical transition reduces the optical opening from Dia. 404 to Dia. 356 over 90 mm height. Its protected silver surface target is >=98% reflectance.

FEATURE	SOURCE VALUE
Top diameter	404 mm
Bottom diameter	356 mm
Height	90 mm
Slant height	93.15 mm
Blank outer radius	783.97 mm
Blank inner radius	690.83 mm
Sector angle	92.76 deg
Overlap tab	15 mm

BUILD CONTROL	WHY
Paper/plastic trial first	Verify assembled diameters before cutting reflective material.
Reflective face	Keep smooth, wrinkle-free, clean; no exposed adhesive or open seam.
Concentricity	Lower opening must remain concentric with the tube collar.
Handling	Use clean gloves/lint-free methods; optical surface defects reduce throughput.

NO WRINKLES / OIL CANNING	Surface waviness or seam defects scatter light. The optical finish of the CPC is a functional requirement, not cosmetic.

9. Clamp, gasket, and thermal movement - assembly logic



Purpose: seal without rigidly locking the polycarbonate flange.

Trial torque 1.5-2.0 N m; verify after leak and thermal cycling.

FEATURE	SOURCE VALUE	PURPOSE
Dome holes	Dia. 7.0	Oversize relative to fastener to allow thermal movement
Clamp ring holes	Dia. 5.5 on PCD Dia. 430	16 positions at 22.5 deg spacing
EPDM gasket	3.0 mm; target 20-25% compression	Continuous weather seal without local hard spots
Compression stop sleeve	7.3-7.6 long through stack	Prevents over-compression / protects polymer flange
Initial torque	1.5-2.0 N m trial value	Verify after leak and thermal-cycle testing
Fastener	M5 x 20 A4 stainless + bonded washer	Avoid direct point load on polycarbonate

THERMAL MOVEMENT	Do not rigidly lock the polycarbonate flange between metal parts. The package deliberately uses oversized polymer holes, bonded washers, and stop sleeves.

10. Components, material compatibility, and substitution control

COMPONENT	MATERIAL	CRITICAL REQUIREMENT
Collector dome	UV optical polycarbonate	5.0 sheet; exterior UV/hard coat; finished wall >=3.0
Prism insert	Optical PC or PMMA	1.0 base; optical adhesive or mechanical retention only after compatibility verification
Clamp ring	5052-H32 or 6061-T6 Al	3.0; anodized/powder coated; flatness <=0.50
Gasket	UV/ozone resistant EPDM	3.0 closed cell; 20-25% compression target
CPC reflector	Aluminum + protected silver film	0.8; reflectance target >=98%; smooth, wrinkle-free
Fasteners	A4 stainless steel	16 x M5 x 20; bonded washer + stop sleeve
Tube collar	Aluminum or galvanized steel	0.8; Dia. 356 ID x 100 engagement

CHEMICAL COMPATIBILITY	Obtain supplier compatibility statements for sealants, tapes, cleaners, adhesives, and coatings used with the selected polycarbonate. Incompatible chemicals can cause crazing or stress cracking.

11. Validation plan - what must be measured before release

TEST	METHOD	INITIAL ACCEPTANCE TARGET
Dimensional	CMM/template/radial height gauge at tabulated meridians	Profile +/-0.50; flange flatness <=0.50; min wall >=3.0
Water leakage	Spray/rain test with gasket and representative roof interface	No water beyond intended drainage/seal plane
Thermal cycle	Representative hot-sun/cold-night; at least 10 cycles	No cracks, crazing, fastener distress, seal displacement, loss of torque control
Impact / hail	Applicable local skylight impact method + safety factors	No penetration/dangerous fragmentation; qualified laboratory certification
Optical throughput	Compare above-dome light with diffuser output	Stable transmission; compare smooth/prism/production-intent assemblies
Directional response	Calibrated source or outdoor altitude/azimuth sweep	Useful low-sun gain without unacceptable glare
Environmental durability	Accelerated UV, humidity, adhesion, coating checks	No unacceptable yellowing, haze, delamination, reflective-film corrosion

INSTRUMENT FIRST ARTICLES	Retain material batch, thickness map, tool temperature, forming temperatures, cooling time, final dimensions, and all test outcomes so design changes can be traced.

12. Failure clues, release gates, and who closes them

SYMPTOM	FIRST CHECKS
Thin apex / transition	Heat distribution, sheet conditioning, draw ratio, forming method; do not trim away evidence before mapping
Flange warped / poor seal	Cooling-on-tool time, trim plane, local chilling, clamp flatness
Cracks/crazing near holes	Drilling technique, point loading, thermal restraint, cleaner/sealant compatibility
Leak at clamp	Gasket continuity/compression, flange flatness, stop sleeves, star tightening
Low directional gain	Prism orientation/geometry, actual resin/wall, CPC alignment/surface, ray-trace assumptions
Haze / optical loss	Tool finish, scratches, contamination, material/coating compatibility

Release gates

ITEM TO FREEZE / RELEASE	GATE	RESPONSIBLE CLOSURE
Dome tool	At least three measured parts show repeatable profile and wall distribution	Fabrication/tooling + engineering review
Prism geometry	Ray tracing correlated with physical directional tests	Optical engineering / test
Clamp torque / stop sleeve	Leakage and thermal-cycle tests completed	Mechanical/weather test
Permanent outdoor use	Structural, weather, fire, impact/hail, safety/code requirements confirmed	Qualified engineer + test lab + AHJ/project team

13. Prototype build record and change-control sign-off

Use this page as an index to the actual evidence. It does not replace laboratory or engineering approval.

FIELD	ENTRY
Prototype ID	_____
Sheet lot / grade	_____
Tool revision	_____
Form date	_____
Starting sheet thickness	_____
Minimum finished wall	_____
Prism revision / orientation	_____
CPC revision	_____

[ ] Profile / flange report attached

[ ] Wall-thickness map attached; min >=3.0 mm

[ ] Optical/visual inspection passed

[ ] Gasket/clamp/stop sleeves verified

[ ] Water leakage test complete

[ ] Thermal cycle complete

[ ] Optical throughput baseline complete

[ ] Directional response test complete

[ ] Material / chemical compatibility evidence retained

[ ] Applicable structural/weather/impact/fire/code approvals closed before permanent use

REVIEW	NAME / SIGNATURE	DATE
Fabrication	_____	_____
Optical / test	_____	_____
Project / installation release	_____	_____

SOURCE PACKAGE FOLLOWS	The original 9-page collector dome engineering package is appended unchanged after this supplement. It remains the controlling source for CAD/profile data, optical geometry, BOM, fabrication instructions, and release limitations.

COMPLETE ENGINEERING DESIGN PACKAGE

Tubular Daylight Collector Dome

Detailed prototype geometry, optical collection principles, mold data, component interfaces, fabrication sequence, and validation plan for a 14-inch nominal tubular daylight device.

DOCUMENT	DESIGN BASIS	STATUS	UNITS
Engineering design description	14-inch / 356 mm clear aperture	Prototype - Rev A	mm unless noted

Design intent

The green collector at the top of the daylight tube is treated as a clear, UV-stabilized optical dome with a directional prismatic sector and a reflective transition. The dome protects the opening, admits diffuse sky light, accepts high-angle direct sunlight, and redirects selected low-angle sunlight toward the reflective tube.

PARAMETER	DESIGN VALUE	PURPOSE
Clear tube aperture	Dia. 356	Matches 14-inch nominal tube
Dome outer base	Dia. 420	Optical entrance and sealing land
Dome rise	230	Broad sky view and useful incidence angles
Mounting flange	Dia. 460	Clamp, gasket, and fastener interface
Nominal sheet	5.0 optical PC	Allows finished wall >= 3.0 after forming
CPC transition	Dia. 404 to 356 x H 90	Guides light into the tube

ENGINEERING LIMIT
This is a preliminary prototype design, not a certified roof product. Wind, snow, hail, impact, fire, flashing, leakage, structural attachment, and local code compliance must be checked by qualified specialists before outdoor installation.

1. Geometry and dimensional definition

The collector is an ellipsoid of revolution rather than a perfect hemisphere. A slightly taller profile improves sky exposure and gives room for the directional prism sector while keeping a broad flange for sealing.

OUTER SURFACE EQUATION

Revolve the profile 360 degrees about the z-axis: $\mathbf{(r / 210)^2 + (z / 230)^2 = 1}$, for 0 <= z <= 230. Here r is radial distance from the axis and z is height above the dome base plane.

Male mold coordinates

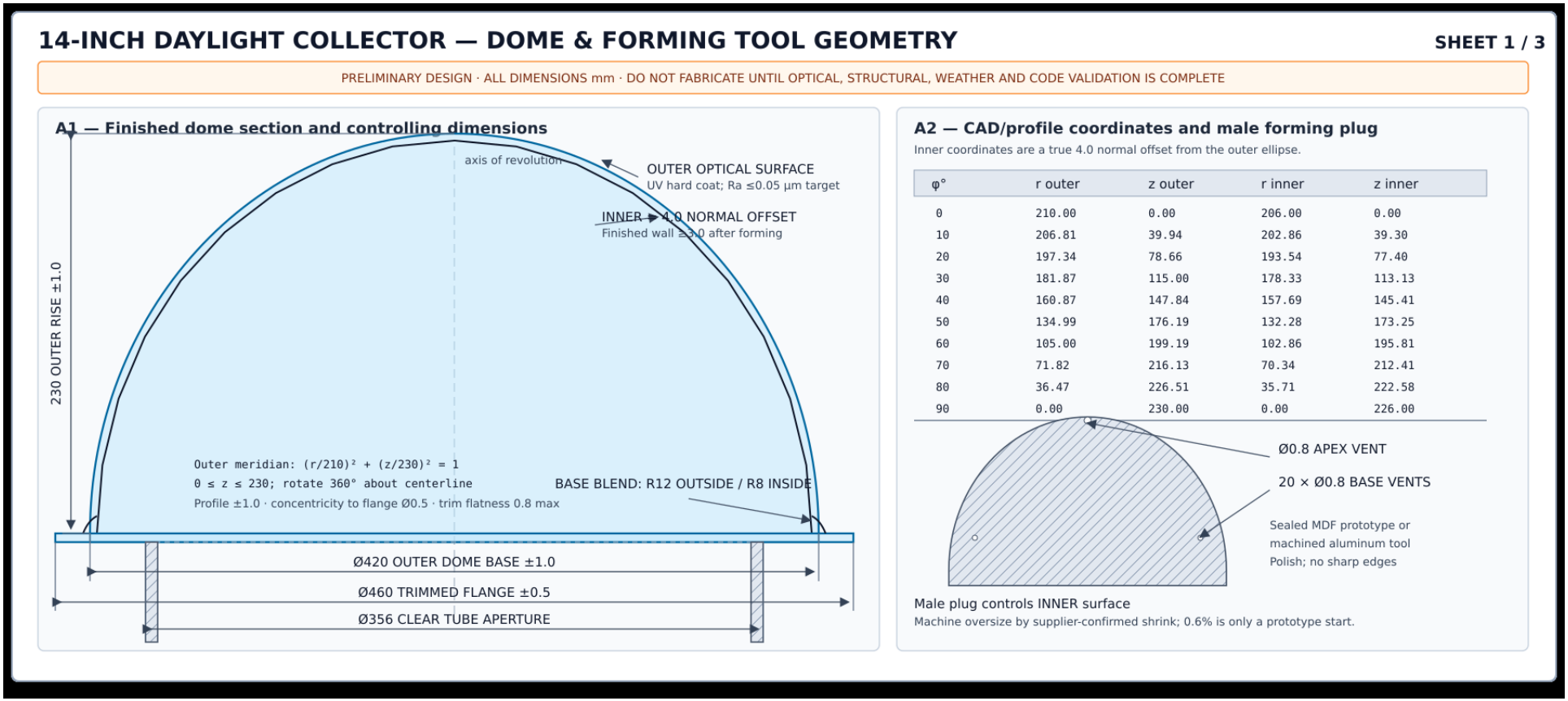
The inner surface is defined as a 4.0 mm normal offset from the outer optical surface. The male plug follows this inner surface. Use the values below to generate a revolved CAD spline; do not simply subtract 4.0 mm in the radial direction.

MERIDIAN ANGLE	OUTER r	OUTER z	INNER r	INNER z
0 deg	210.00	0.00	206.00	0.00
10 deg	206.81	39.94	202.86	39.30
20 deg	197.34	78.66	193.54	77.40
30 deg	181.87	115.00	178.33	113.13
40 deg	160.87	147.84	157.69	145.41
50 deg	134.99	176.19	132.28	173.25
60 deg	105.00	199.19	102.86	195.81
70 deg	71.82	216.13	70.34	212.41
80 deg	36.47	226.51	35.71	222.58
90 deg	0.00	230.00	0.00	226.00

Tool and finished-part tolerances

- Profile tolerance: +/-0.50 mm over the optical surface; no local waviness greater than 0.25 mm over a 50 mm chord.
- Base cut plane: flat within 0.50 mm; flange outer diameter Dia. 460 +/-0.50 mm.
- Finished wall: target 4.0 mm, minimum 3.0 mm at the apex and transition regions.
- Optical finish: polished tool surface; remove machining marks before forming. Do not flame-polish polycarbonate.
- Prototype shrink allowance: begin with 0.6 percent only as a trial value; confirm the actual resin/forming process and compensate the tool after measurement.

Overall dome geometry, mold section, and mounting interface



Sheet 1 establishes the ellipsoidal profile, clear aperture, flange, 4 mm normal wall definition, hole pattern, gasket, and clamp stack.

2. Optical collection design

The collector combines three optical functions: a smooth transparent dome for broad-angle transmission, a directional prism sector for low-angle sunlight, and a reflective compound parabolic concentrator (CPC) transition that guides admitted rays into the tube.

ELEMENT	STARTING DESIGN	OPTICAL ROLE
Clear dome	UV-stabilized optical polycarbonate; exterior UV and hard coat	Transmit diffuse sky light and direct sunlight over a wide incidence range
Directional prism sector	120-degree azimuth sector; meridian band 5 to 55 degrees	Refract low-altitude sunlight downward instead of allowing it to cross the dome and escape
Prism geometry	2.50 pitch; 1.17 depth; 35 / 55 degree facets; 1.0 base film	Starting geometry for ray tracing and prototype trials
CPC transition	Dia. 404 top, Dia. 356 bottom, height 90; protected silver reflectance target >= 98 percent	Capture edge rays and couple them to the reflective tube

Orientation

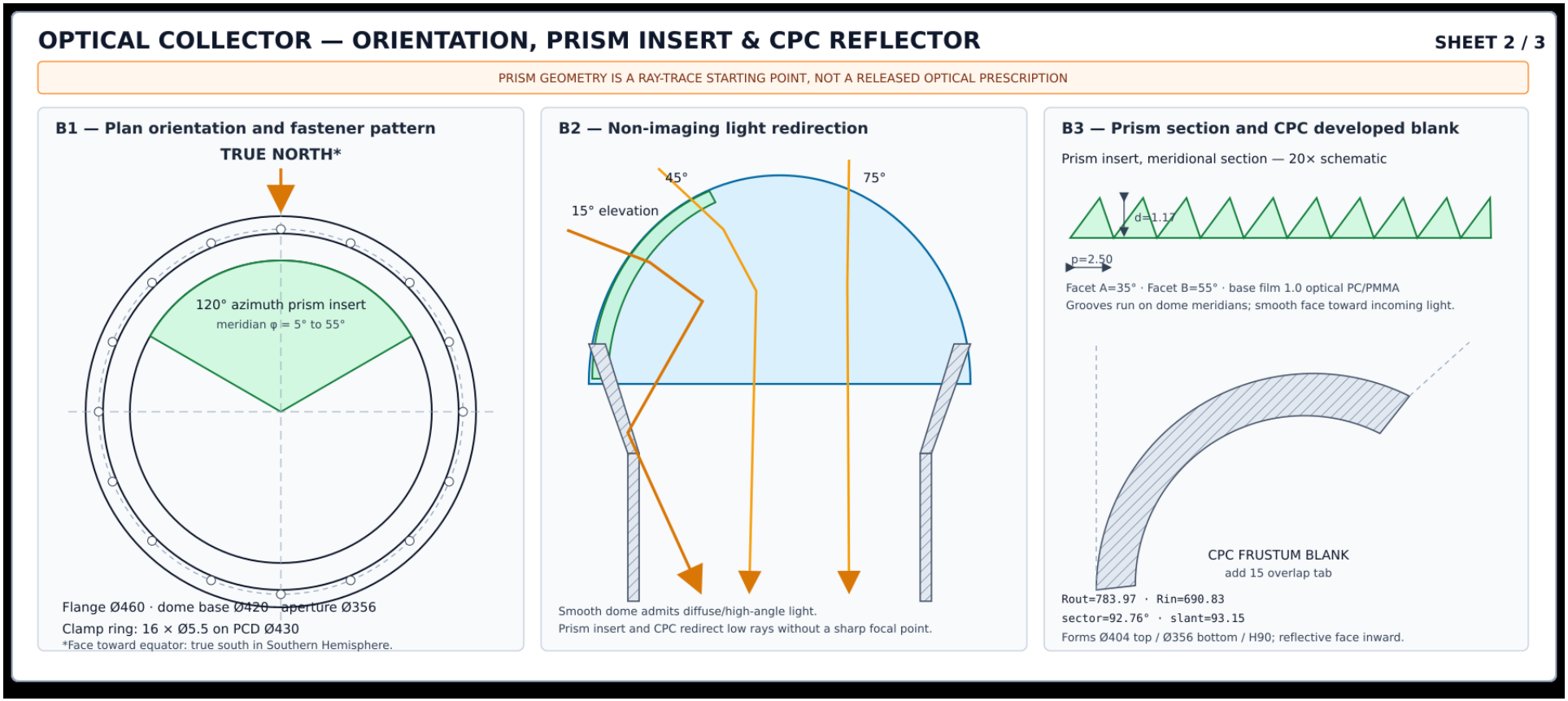
Place the prismatic sector toward the equator: face it generally south in the Northern Hemisphere and north in the Southern Hemisphere. Mark the equator-facing direction permanently on the clamp ring. Final azimuth should be chosen using the site latitude, roof pitch, surrounding shading, and the hours when useful daylight is most valuable.

Optical principles

- Snell refraction at the dome surface changes ray direction according to the refractive indices of air and polycarbonate. Use the actual supplier dispersion data in the ray model.
- Smooth, gradual curvature reduces sharp incidence transitions and supports daylight acceptance from a large portion of the sky vault.
- The asymmetric prism facets provide a preferred bend direction. The 35 / 55 degree facet pair is a manufacturable first trial, not a final prescription.
- The reflective transition must avoid wrinkles, oil canning, exposed adhesive, or open seams because each defect scatters light and reduces throughput.
- Evaluate annual useful lumens at the diffuser, not only noon peak illuminance. A wider, more uniform acceptance band is usually preferable to a narrow peak.

RAY-TRACE REQUIRED
Optimize the prism pitch, facet angles, sector width, and meridian coverage using the final resin index, wall thickness, latitude, weather data, tube reflectance, diffuser distribution, and roof orientation. The dimensions shown are a controlled prototype starting point.

Optical ray concepts, prism insert, CPC transition, and developed blank



Sheet 2 shows the directional daylight strategy and starting dimensions for the prismatic sector and reflective transition.

3. Components and interfaces

Use optical-grade, outdoor-qualified materials and keep all sealants, tapes, cleaners, and coatings chemically compatible with the selected polycarbonate grade. Obtain written compatibility statements from suppliers.

ITEM	COMPONENT	MATERIAL / GRADE	QTY	SIZE	NOTES
1	Collector dome	UV optical polycarbonate	1	5.0 sheet	Exterior UV and hard coat; clear; finished wall >= 3.0
2	Prism insert	Optical PC or PMMA	1	1.0 base	Embossed or molded; bond only with verified optical adhesive or mechanical retention
3	Clamp ring	5052-H32 or 6061-T6 Al	1	3.0	Anodized or powder coated; flatness <= 0.50
4	Gasket	UV and ozone resistant EPDM	1	3.0	Closed cell; design for 20 to 25 percent compression
5	CPC reflector	Aluminum + protected silver film	1	0.8	Reflectance target >= 98 percent; smooth, wrinkle-free optical surface
6	Fasteners	A4 stainless steel	16	M5 x 20	Bonded washer plus compression stop sleeve; avoid point loading
7	Tube collar	Aluminum or galvanized steel	1	0.8	Dia. 356 ID x 100 engagement; seal to tube system

Fastener and seal design

FEATURE	DOME	CLAMP RING	DESIGN CONTROL
Hole diameter	Dia. 7.0	Dia. 5.5	Allows dome thermal movement around M5 fastener
Bolt circle	PCD Dia. 430	PCD Dia. 430	16 positions, equally spaced at 22.5 degrees
Compression sleeve	7.3 to 7.6 long	Through ring/gasket stack	Stops over-compression and protects the polymer flange
Initial torque	Not torque-bearing	1.5 to 2.0 N m	Trial value; verify after leak and thermal cycling
Gasket compression	Uniform flange support	Continuous clamp	Target 20 to 25 percent; no local hard spots

THERMAL MOVEMENT
Do not rigidly lock the polycarbonate flange between metal parts. Oversize polymer holes, use bonded washers and stop sleeves, and allow the flange to expand without stress concentration.

ENGINEERING DRAWING - SHEET 3 OF 3

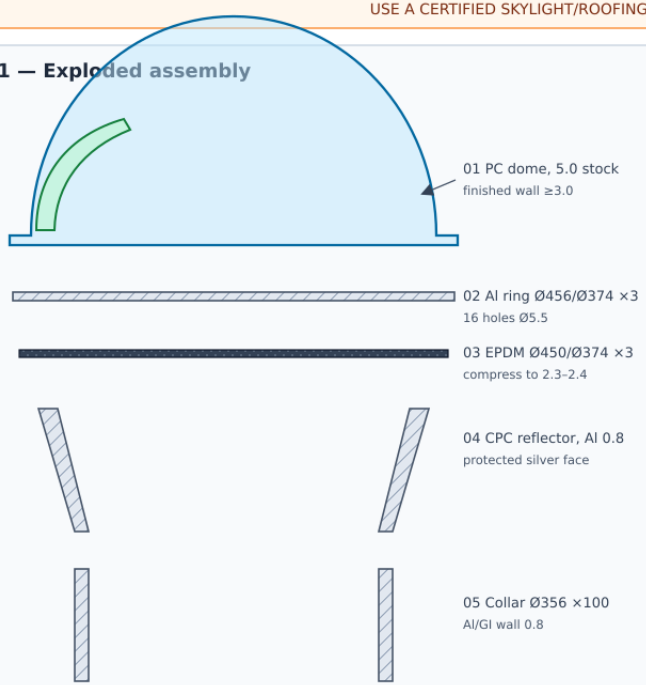
Small-batch fabrication sequence, assembly stack, and inspection controls

PARTS, SEALING DETAIL & PROTOTYPE FABRICATION SEQUENCE

SHEET 3 / 3

USE A CERTIFIED SKYLIGHT/ROOFING ENGINEER FOR WIND, SNOW, IMPACT, FIRE, FLASHING AND WATER-PENETRATION REQUIREMENTS

C1 — Exploded assembly

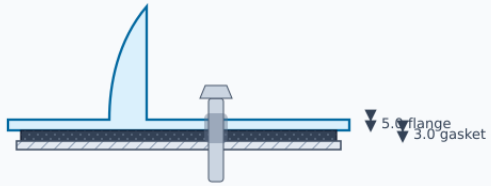


06 · 16× M5×20 A4-70 screws, bonded washers, stop sleeves and locknuts

C2 — Prototype bill of materials

Item	Qty	Material / grade	Finish	Critical requirement
01 Dome	1	UV optical PC sheet, 5.0	Exterior UV/hard coat	Clear; t min 3.0
02 Prism	1	Optical PC/PMMA, 1.0 base	Embossed prisms	Orientation
03 Ring	1	5052-H32 or 6061-T6 Al, 3.0	Anodize/powder coat	Flatness ≤0.5
04 Gasket	1	UV/ozone EPDM, 3.0	Closed cell	20–25% compression
05 CPC	1	Al sheet, 0.8	Protected silver ≥98% R	No wrinkles
06 Screws	16	M5×20 A4 stainless	Washer + stop sleeve	No point loading

C3 — Weather seal and thermal-expansion detail



Dome hole Ø7.0; clamp-ring hole Ø5.5
EPDM sleeve/washer centers M5 screw
Stop sleeve 7.3–7.6 controls gasket compression
Initial torque 1.5–2.0 N·m; verify after thermal cycle

C4 — Small-batch fabrication sequence

1 · Machine/polish male plug from profile table
2 · Predry PC per grade; start 120°C, 2–3 h
3 · Pressure/vacuum form in supplier window
4 · Cool on tool; trim; drill oversize holes
5 · Fit prism, CPC, gasket and clamp ring
6 · Leak, impact, load, thermal and ray tests

Never flame-polish PC; avoid incompatible solvent adhesives.

DESIGN BASIS: nominal 356 mm tube; northern-hemisphere orientation shown. Match materials and processing to the selected grade, latitude, roof pitch, climate loads and applicable skylight standards.

Sheet 3 is the workshop reference for plug tooling, thermoforming, trimming, prism and CPC installation, sealing, and prototype checks.

4. Prototype fabrication and assembly

The process below is intended for small-batch development. Confirm all temperature limits with the exact sheet grade and forming equipment; supplier values override generic starting conditions.

STEP	OPERATION	DETAIL / CONTROL
1	Build male plug	Machine sealed MDF for initial trials or aluminum for repeat production. Use the inner profile coordinate table and include a stable flange land.
2	Finish the tool	Seal and polish the optical surface. Provide one Dia. 0.8 apex vent and twenty Dia. 0.8 base vents, deburred and connected to vacuum.
3	Condition sheet	Keep protective masking in place where allowed. Predry the selected PC grade per supplier instructions; 120 C for 2 to 3 hours is only a starting trial.
4	Thermoform	Pressure form or vacuum form in the supplier process window. Heat uniformly, avoid chilling at the clamp, and record sheet temperature and draw ratio.
5	Cool on tool	Hold until the part is dimensionally stable. Premature release can cause springback, warpage, and flange distortion.
6	Trim and drill	Trim flange to Dia. 460. Drill Dia. 7.0 polymer holes on PCD Dia. 430 using sharp tools and backed support. Deburr without solvent.
7	Install optics	Fit the directional prism sector, then form and install the CPC. Keep optical surfaces clean and avoid trapped dust or adhesive squeeze-out.
8	Seal and clamp	Place EPDM gasket, clamp ring, stop sleeves, bonded washers, and M5 fasteners. Tighten gradually in a star pattern to 1.5 to 2.0 N m.

CPC developed blank

For the Dia. 404 to Dia. 356 by 90 mm conical transition, use a slant height of 93.15 mm. The starting annular-sector blank is: outer radius 783.97 mm, inner radius 690.83 mm, sector angle 92.76 degrees, plus a 15 mm overlap tab. Cut a paper or thin plastic trial first and verify the assembled diameters before cutting reflective material.

Assembly inspection

- Confirm equator-facing prism orientation and mark it on the clamp ring.
- Check the gasket is continuous, unwrinkled, and compressed evenly around the circumference.
- Confirm no fastener head or washer applies point load directly to the polycarbonate.
- Verify CPC seams are closed, the reflective face is undamaged, and the lower opening remains concentric with the tube.
- Use only lint-free cleaning materials and a cleaner approved by the sheet manufacturer.

5. Prototype validation plan

Treat the first articles as instrumented prototypes. Record material batch, sheet thickness map, tool temperature, forming temperatures, cooling time, final dimensions, and all test outcomes so the design can be corrected before release.

TEST	METHOD	INITIAL ACCEPTANCE TARGET
Dimensional inspection	CMM, template, or radial height gauge at the tabulated meridians	Profile within +/-0.50; flange flatness <= 0.50; minimum wall >= 3.0
Water leakage	Spray/rain test with installed gasket and representative roof interface	No water beyond the designed drainage/seal plane
Thermal cycle	Cycle representative hot-sun and cold-night temperatures; inspect at least 10 cycles	No cracks, crazing, fastener distress, seal displacement, or loss of torque control
Impact / hail screening	Use the applicable local skylight impact method and safety factors	No penetration or dangerous fragmentation; certify through a qualified laboratory
Optical throughput	Measure outdoor illuminance or luminous flux above the dome and at the diffuser	Stable transmission; compare smooth dome, prism prototype, and production-intent assembly
Directional response	Rotate a calibrated collimated source or test outdoors across solar altitude and azimuth	Prism sector increases useful tube-coupled light in the intended low-sun range without unacceptable glare
Environmental durability	Accelerated UV, humidity, adhesion, and coating checks using supplier and local standards	No unacceptable yellowing, haze, delamination, or reflective-film corrosion

Release gates

- Freeze the dome tool only after at least three measured parts show repeatable profile and wall distribution.
- Freeze prism geometry only after ray tracing is correlated with physical directional-light tests.
- Freeze clamp torque and stop-sleeve length only after leakage and thermal-cycle tests.
- Do not sell or permanently roof-mount the unit until a qualified engineer and test laboratory confirm the applicable structural, weather, fire, and safety requirements.

References and technical basis

- VELUX Commercial. Tubular daylighting sizing guidance. https://commercial.veluxusa.com/products/tubular/commercial-suntunnel/sizing
- POLYVANTIS. LEXAN solid polycarbonate sheet handling and processing resources. https://www.polyvantis.com/en/products/handling-care/polycarbonate/lexan-sheet/lexan-solid-sheet
- Covestro. Makrolon polycarbonate product information. https://solutions.covestro.com/en/brands/makrolon
- US Patent 5,655,339. Prismatic tubular skylight principle. https://patents.google.com/patent/US5655339A/en
- EP Patent 1,931,837 B1. Variable prism daylight-collection principle. https://patents.google.com/patent/EP1931837B1/en
- VELUX Commercial. Tubular daylighting technical resources. https://commercial.veluxusa.com/resources/tubular

FINAL RESPONSIBILITY

A roof-mounted daylight collector is a building-envelope and overhead safety component. The fabricator and installer remain responsible for material traceability, optical performance, structural design, drainage, flashing, sealing, fall protection, maintenance access, and all applicable approvals.